Food Delivery Customer Churn Prediction

Objective

The objective of this project is to predict which customers are likely to stop ordering food and identify key factors influencing churn. This helps businesses design targeted retention strategies.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading Dataset

```
df = pd.read_csv("/content/onlinefoods.csv")
df.head()
```

	Age	Gender	Marital Status	Occupation	Monthly Income	Educational Qualifications	Family size	latitud
0	20	Female	Single	Student	No Income	Post Graduate	4	12.976
1	24	Female	Single	Student	Below Rs.10000	Graduate	3	12.977
2	22	Male	Single	Student	Below Rs.10000	Post Graduate	3	12.955
3	22	Female	Single	Student	No Income	Graduate	6	12.947
4	22	Male	Single	Student	Below Rs.10000	Post Graduate	4	12.985

Next steps: Generate code with df | New interactive sheet

Data Overview

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 388 entries, 0 to 387
Data columns (total 13 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Age                         388 non-null    int64
 1   Gender                      388 non-null    object
 2   Marital Status              388 non-null    object
 3   Occupation                  388 non-null    object
 4   Monthly Income              388 non-null    object
 5   Educational Qualifications  388 non-null    object
 6   Family size                 388 non-null    int64
 7   latitude                    388 non-null    float64
 8   longitude                   388 non-null    float64
 9   Pin code                    388 non-null    int64
 10  Output                      388 non-null    object
 11  Feedback                    388 non-null    object
 12  Unnamed: 12                 388 non-null    object
dtypes: float64(2), int64(3), object(8)
memory usage: 39.5+ KB
```

	Age	Family size	latitude	longitude	Pin code
count	388.000000	388.000000	388.000000	388.000000	388.000000
mean	24.628866	3.280928	12.972058	77.600160	560040.113402
std	2.975593	1.351025	0.044489	0.051354	31.399609
min	18.000000	1.000000	12.865200	77.484200	560001.000000
25%	23.000000	2.000000	12.936900	77.565275	560010.750000
50%	24.000000	3.000000	12.977000	77.592100	560033.500000
75%	26.000000	4.000000	12.997025	77.630900	560068.000000
max	33.000000	6.000000	13.102000	77.758200	560109.000000

Creating Target Variable (Churn)

```
df["Churn"] = (df["Output"] == "No").astype(int)
```

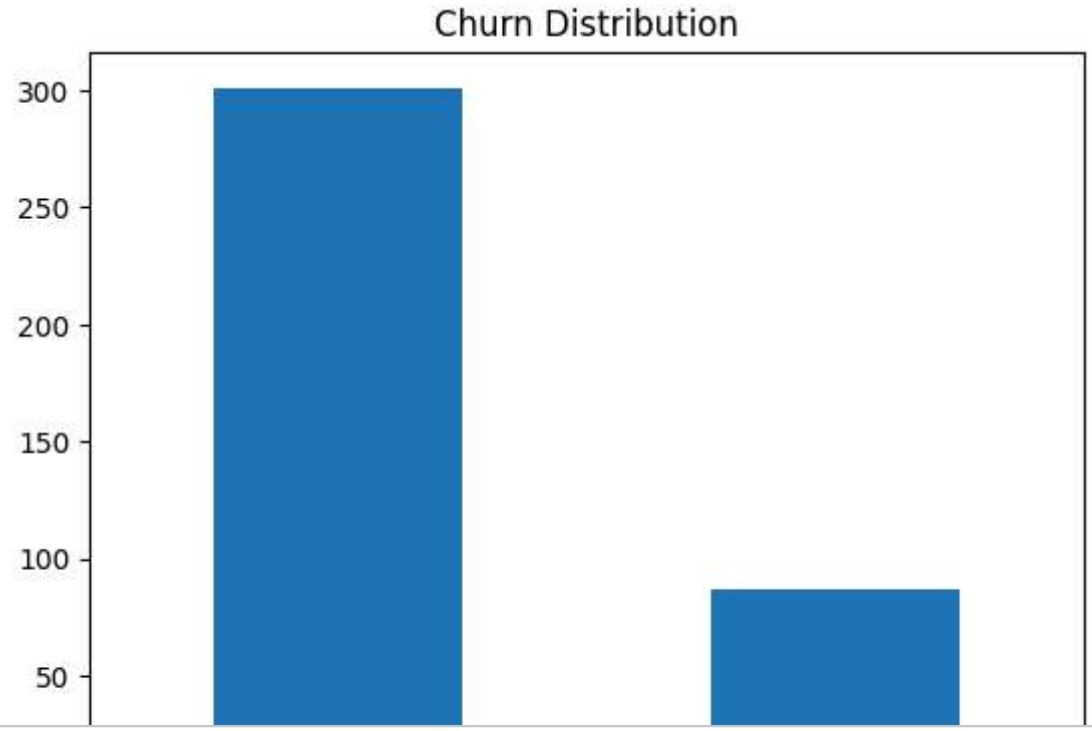
Data Cleaning

```
df.dropna(inplace=True)
```

Exploratory Data Analysis (EDA)

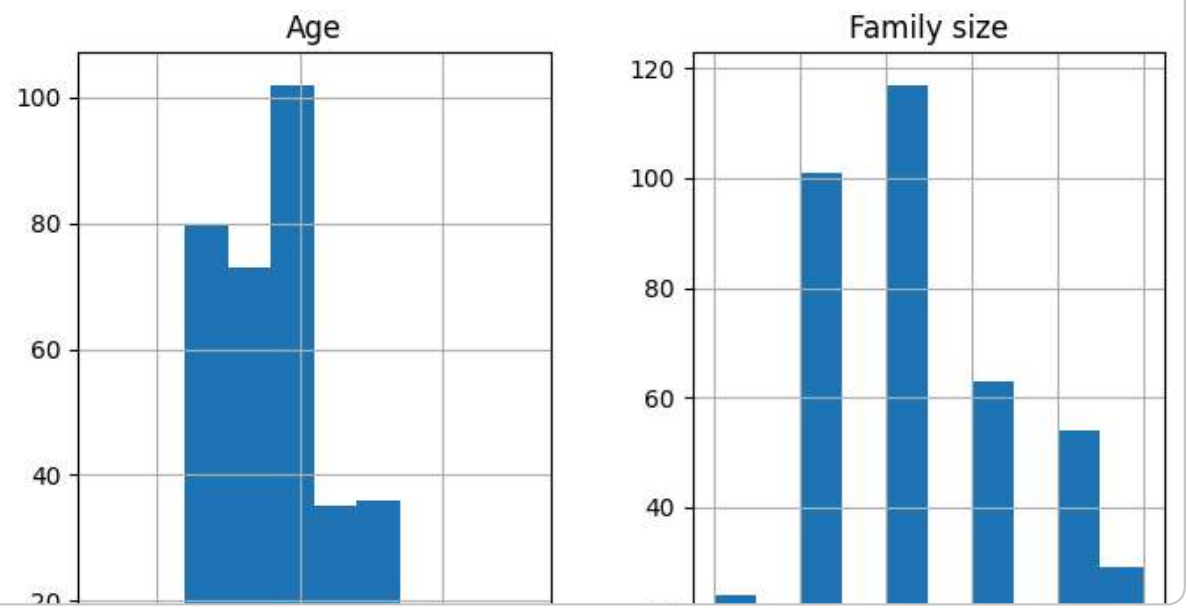
Churn Distribution

```
df["Churn"].value_counts().plot(kind='bar')
plt.title("Churn Distribution")
plt.show()
```



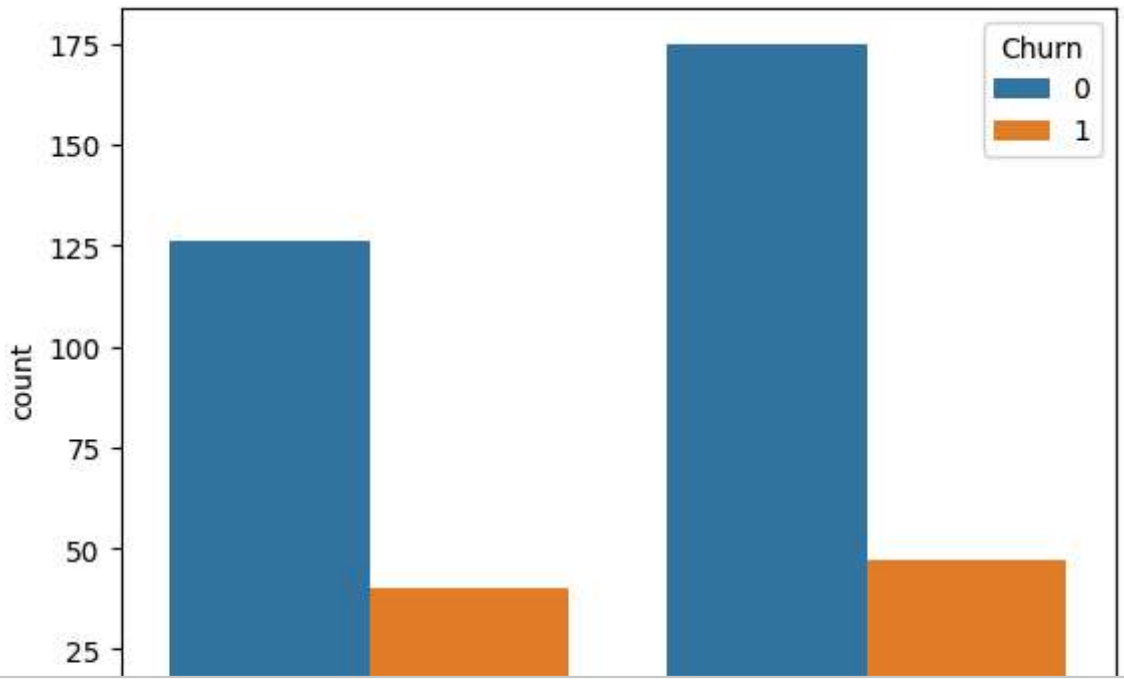
Numerical Distribution

```
df[["Age","Family size"]].hist(figsize=(8,5))
plt.show()
```

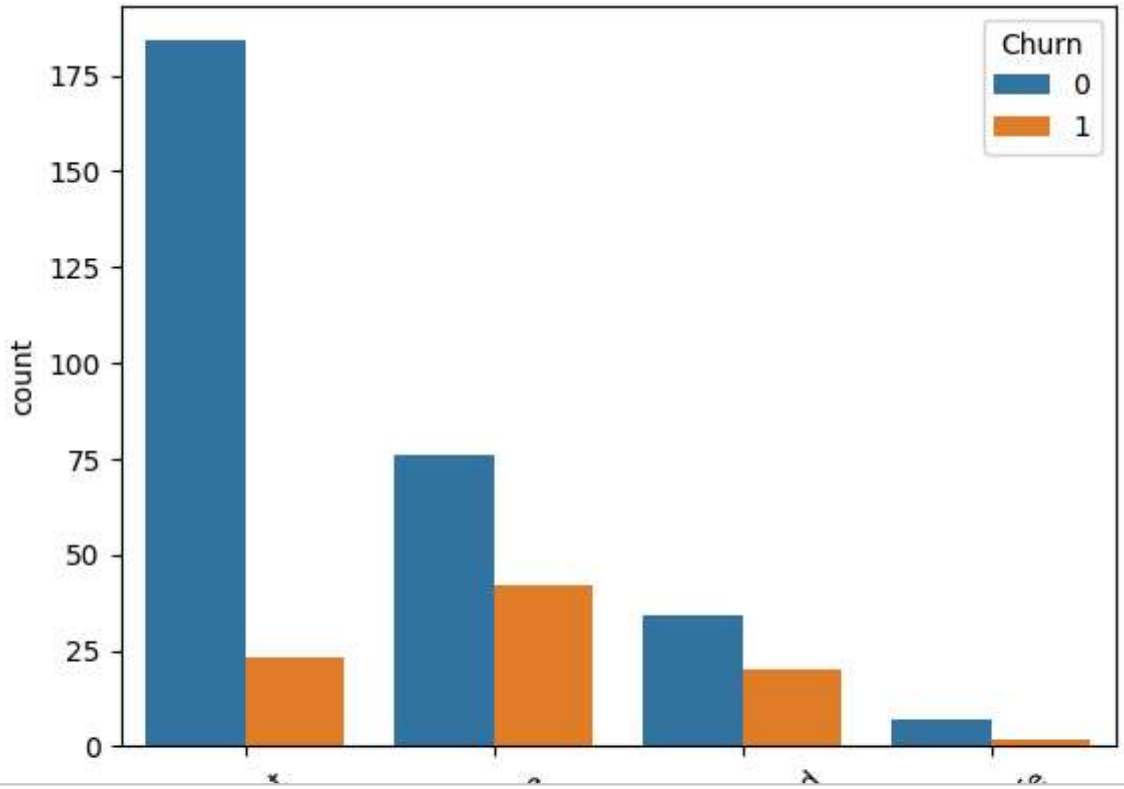


Categorical Analysis

```
sns.countplot(x="Gender", hue="Churn", data=df)
plt.show()
```

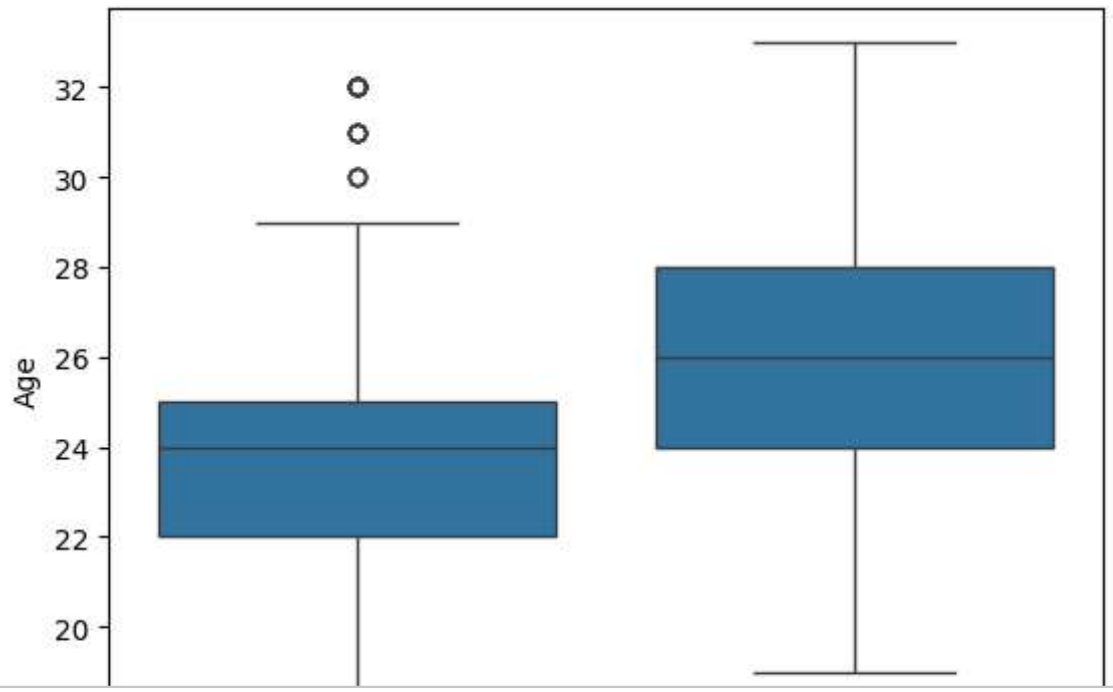



```
sns.countplot(x="Occupation", hue="Churn", data=df)
plt.xticks(rotation=45)
plt.show()
```

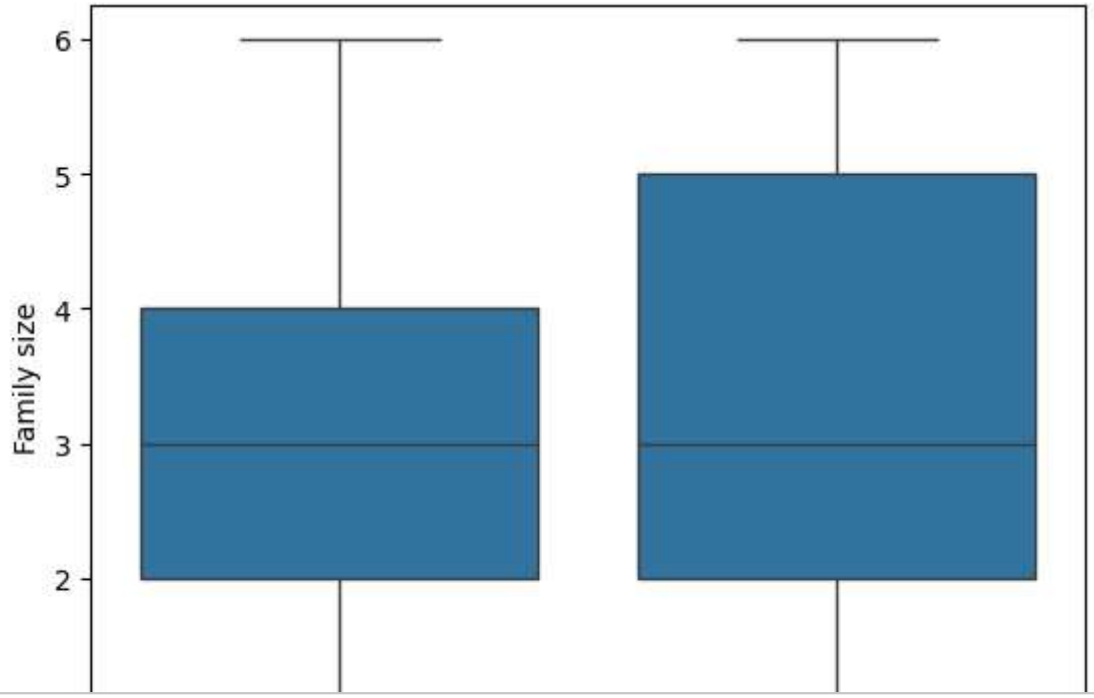


Relationship Analysis

```
sns.boxplot(x="Churn", y="Age", data=df)
plt.show()
```



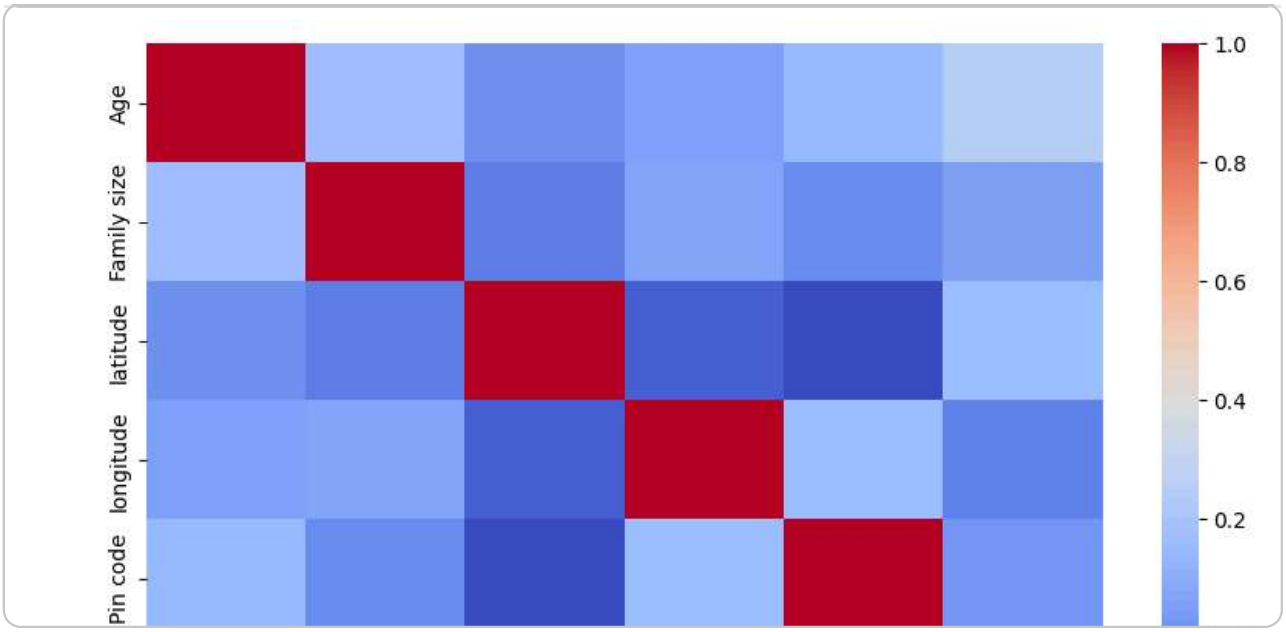

```
sns.boxplot(x="Churn", y="Family size", data=df)
plt.show()
```



Correlation

```
numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), cmap="coolwarm")
plt.show()
```



EDA Insights

- Younger users show higher churn tendency
- Certain occupations have higher churn rates
- Family size influences ordering behavior
- Moderate class imbalance observed

Feature Engineering

```
df["Engagement"] = df["Family size"] * df["Age"]
```

```
df["Age_Group"] = pd.cut(df["Age"], bins=[18,25,35,50,70], labels=["Young","
```

```
df["High_Value"] = (df["Family size"] > df["Family size"].median()).astype(i
```

```
df["Risk_Score"] = df["Age"] / (df["Family size"] + 1)
```

Encoding Categorical Variables

```
df = pd.get_dummies(df, drop_first=True)
```

Feature Selection

```
X = df.drop("Churn", axis=1)
y = df["Churn"]
```

Class Imbalance

```
y.value_counts()
```

Churn	count
0	301
1	87

dtype: int64

Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Logistic Regression (Baseline Model)

```
from sklearn.linear_model import LogisticRegression

lr = LogisticRegression()
lr.fit(X_train, y_train)

y_pred = lr.predict(X_test)
```

Model Evaluation

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
[[66  0]
 [ 0 12]]
              precision    recall  f1-score   support
```

```
           0       1.00      1.00      1.00        66
           1       1.00      1.00      1.00        12

    accuracy                           1.00        78
   macro avg       1.00      1.00      1.00        78
weighted avg       1.00      1.00      1.00        78
```

Random Forest Model

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=200,
    max_depth=6,
    class_weight="balanced",
    random_state=42
)

rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
Accuracy: 1.0
```

Model Evaluation

```
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
```

```
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        66
           1       1.00      1.00      1.00        12

    accuracy                           1.00        78
   macro avg       1.00      1.00      1.00        78
weighted avg       1.00      1.00      1.00        78
```

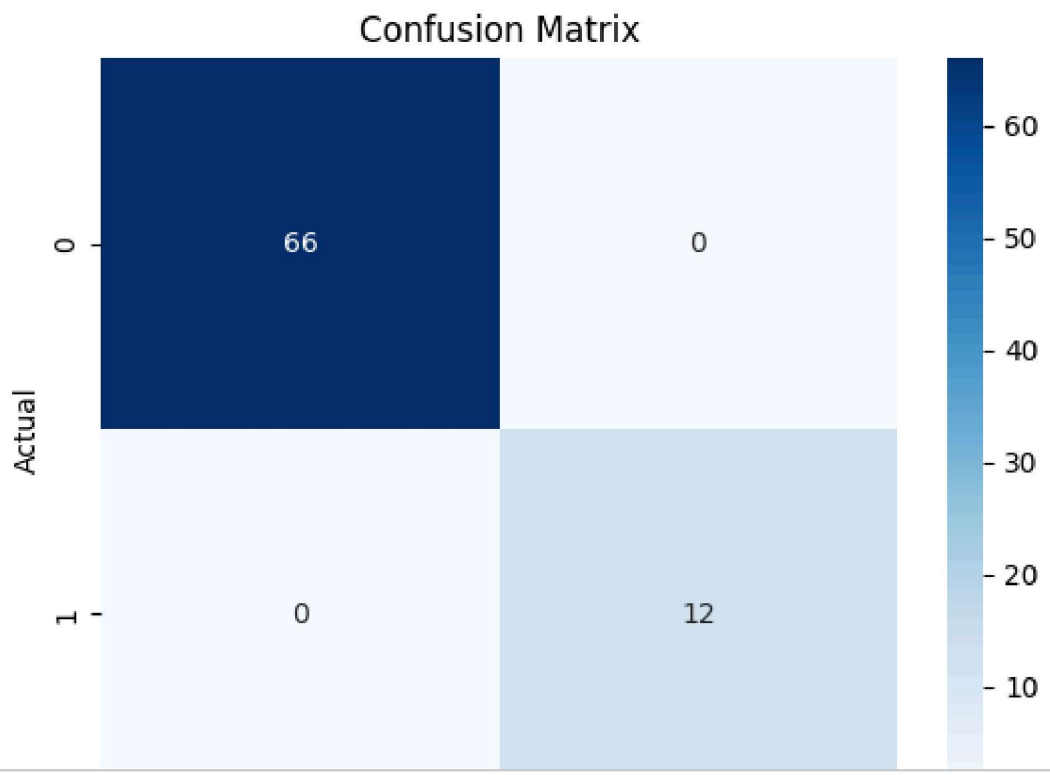
Confusion Matrix

```
from sklearn.metrics import confusion_matrix
import seaborn as sns

cm = confusion_matrix(y_test, y_pred_rf)

sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



Churn Probability Prediction

```
probs = rf.predict_proba(X_test)[:,1]
```

Probability-Based Insight

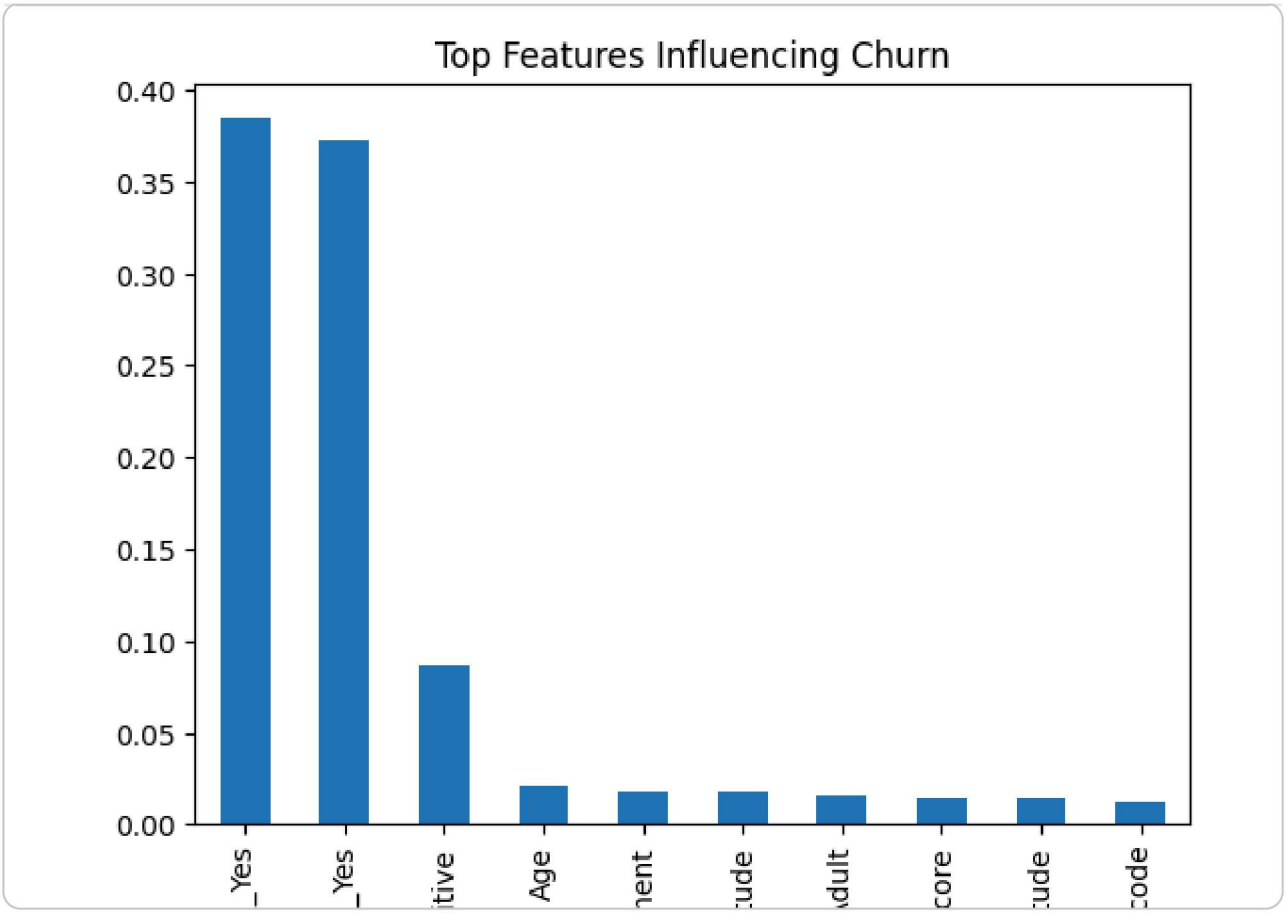
Instead of only predicting churn, the model estimates churn probability, allowing businesses to identify high-risk customers and take preventive action.

Feature Importance

```
importances = rf.feature_importances_

feat_imp = pd.Series(importances, index=X.columns)
feat_imp = feat_imp.sort_values(ascending=False)

feat_imp.head(10).plot(kind='bar')
plt.title("Top Features Influencing Churn")
plt.show()
```



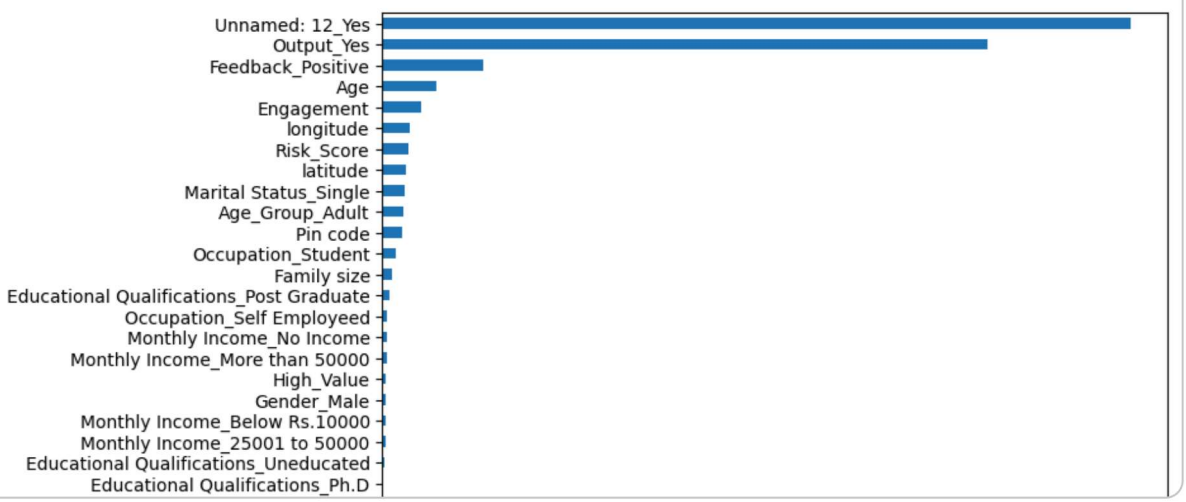
Feature Importance Insights

- Engagement and usage patterns strongly influence churn
- Demographic features also impact behavior
- Low engagement is a key indicator of churn

Feature Importance

```
importances = rf.feature_importances_

feat_imp = pd.Series(importances, index=X.columns)
feat_imp.sort_values().plot(kind='barh', figsize=(8,6)
plt.show()
```



Business Insights

- Users with low engagement are more likely to churn
- Age and family size influence ordering behavior
- High-value users require retention strategies

Business Recommendations

- Target high-risk users with personalized offers
- Improve engagement for low-activity users
- Provide loyalty rewards to retain valuable customers

Conclusion

This project demonstrates how machine learning can be used to predict customer